

**Topic: Magnetic Field, Biot–Savart Law,
Ampere’s law, Magnetic Force on a Moving
Charged Particle, Magnetic lines of induction,
Faraday’s law, Lenz’s Law**

Magnetic Field

A magnetic field is a **vector field** that describes the influence of magnetic forces in a region of space. It is a fundamental concept in physics, primarily associated with moving electric charges, magnetic materials, and time-varying electric fields.

Key Properties of a Magnetic Field

Representation

- Denoted as **B**, the magnetic field is a vector field with both magnitude and direction.

SI Unit

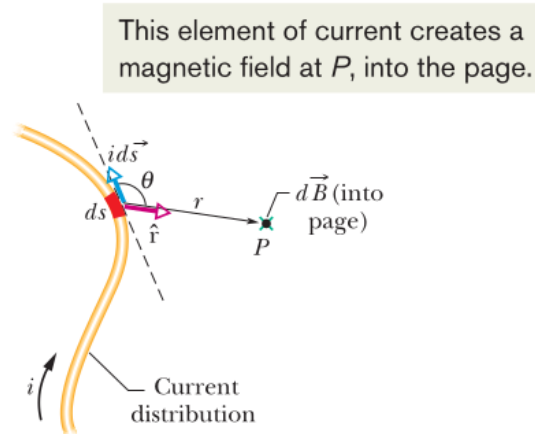
- The SI unit of the magnetic field is the **tesla (T)**.
- Smaller units like the **millitesla (mT)** or **gauss (G)** are also commonly used, where:

$$1 \text{ G} = 10^{-4} \text{ T}$$

Sources of Magnetic Field

1. **Moving Charges:** A moving charge creates a magnetic field around it.
2. **Electric Current:** Current-carrying conductors generate magnetic fields.
3. **Permanent Magnets:** Magnetic materials have intrinsic magnetic fields.
4. **Changing Electric Field:** A time-varying electric field produces a magnetic field.

Biot–Savart Law :



The magnetic field set up by a current-carrying conductor can be determined using the Biot–Savart law. This law states that the contribution $d\mathbf{B}$ to the magnetic field produced by a small current element $i d\mathbf{s}$ at a point P , located a distance r from the current element, is given by:

$$d\mathbf{B} = \frac{\mu_0}{4\pi} \frac{i d\mathbf{s} \times \hat{\mathbf{r}}}{r^2}$$

where:

- μ_0 : Permeability constant ($\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$).
- i : Current through the conductor.
- $d\mathbf{s}$: A small length element of the conductor in the direction of the current.
- r : Distance from the current element to the observation point P .
- $\hat{\mathbf{r}}$: Unit vector pointing from the current element toward P .

The permeability constant, μ_0 , determines the strength of the magnetic field produced by currents and has the value:

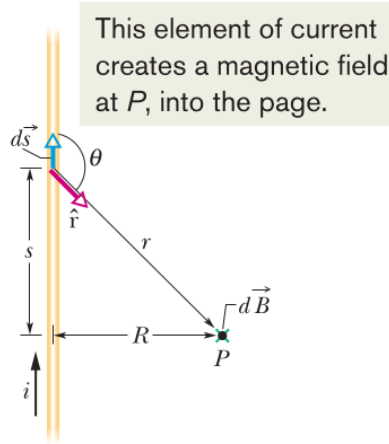
$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A} = 1.26 \times 10^{-6} \text{ T}\cdot\text{m/A}$$

Significance of the Biot–Savart Law

The Biot–Savart law is fundamental for understanding the relationship between electric currents and the magnetic fields they produce. It serves as the basis for calculating the magnetic field due to various current configurations, such as straight wires, circular loops, and solenoids.

Magnetic Field Due to an Infinitely Long Straight Current-Carrying Wire

Consider an infinitely long straight wire carrying a steady current I . We aim to find the magnetic field at a point P at a perpendicular distance R from the wire.



The magnitude of the differential magnetic field produced at P by the current-length element $I ds$ located a distance r from P is given by the Biot-Savart law:

$$dB = \frac{\mu_0 I}{4\pi} \frac{ds \sin \theta}{r^2}$$

For an infinitesimal element ds of the wire, we write:

$$r = \sqrt{s^2 + R^2}, \quad \sin \theta = \sin(\pi - \theta) = \frac{R}{\sqrt{s^2 + R^2}}$$

Using the Biot-Savart law:

$$dB = \frac{\mu_0 I}{4\pi} \frac{R ds}{(s^2 + R^2)^{3/2}}$$

Integrating from $s = -\infty$ to $s = \infty$:

$$B = \frac{\mu_0 I}{4\pi} \int_{-\infty}^{\infty} \frac{R ds}{(s^2 + R^2)^{3/2}}$$

Using the substitution $s = R \tan \theta$, where $ds = R \sec^2 \theta d\theta$, we change the limits:

$$s = -\infty \Rightarrow \theta = -\frac{\pi}{2}, \quad s = \infty \Rightarrow \theta = \frac{\pi}{2}$$

Rewriting the integral:

$$\int_{-\infty}^{\infty} \frac{R ds}{(s^2 + R^2)^{3/2}} = \int_{-\pi/2}^{\pi/2} \frac{R^2 \sec^2 \theta d\theta}{(R^2 \sec^2 \theta)^{3/2}}$$

Simplifying:

$$\int_{-\pi/2}^{\pi/2} \frac{R^2 \sec^2 \theta d\theta}{R^3 \sec^3 \theta} = \int_{-\pi/2}^{\pi/2} \frac{d\theta}{R \sec \theta}$$

Since $\frac{1}{\sec \theta} = \cos \theta$:

$$\int_{-\pi/2}^{\pi/2} \cos \theta d\theta = \sin \theta \Big|_{-\pi/2}^{\pi/2}$$

Evaluating:

$$\sin \left(\frac{\pi}{2} \right) - \sin \left(-\frac{\pi}{2} \right) = 1 - (-1) = 2$$

Substituting back:

$$B = \frac{\mu_0 I}{4\pi} \cdot \frac{2}{R} = \frac{\mu_0 I}{2\pi R}$$

This result shows that the magnetic field forms concentric circles around the wire, with magnitude inversely proportional to the distance R .

Ampère's Law

Ampère's Law relates the magnetic field $\mathbf{B}$ around a closed loop to the total current I_{enc} passing through the loop. It is one of Maxwell's equations and is written in two forms: the integral form and the differential form.

Integral Form

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$$

where:

- $\oint \mathbf{B} \cdot d\mathbf{l}$ is the line integral of the magnetic field $\mathbf{B}$ around a closed loop.
- μ_0 is the permeability of free space, $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$.
- I_{enc} is the net current enclosed by the loop.

Application of Ampère's Law:

Magnetic Field Outside a Long Straight Wire with Current :

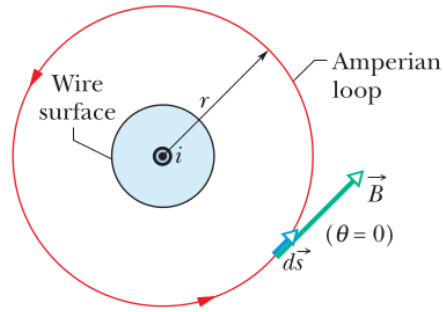


Figure shows a long straight wire that carries current i directly out of the page. The magnetic field B produced by the current has the same magnitude at all points that are the same distance r from the wire; that is, the field B has cylindrical symmetry about the wire. We can take advantage of that symmetry to simplify the integral in Ampere's law if we encircle the wire with a concentric circular Amperian loop of radius r , as in Figure.

Since B is tangent to the loop at every point along the loop, and the angle between ds and B is 0° , we simplify Ampere's law:

$$\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 i_{\text{enc}}$$

$$\oint B ds \cos\theta = \mu_0 i_{\text{enc}}$$

Since B is constant along the loop,

$$B \oint ds = B(2\pi r) = \mu_0 i$$

Solving for B :

$$B = \frac{\mu_0 i}{2\pi r}$$

This confirms that the magnetic field outside a long straight wire is inversely proportional to the radial distance r .

Magnetic Field Inside a Long Straight Wire with Current:

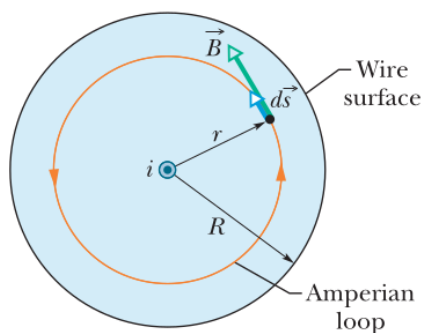


Figure shows the cross-section of a long straight wire of radius R that carries a uniformly distributed current i directly out of the page. Because the current is uniformly distributed over a cross-section of the wire, the magnetic field B must be cylindrically symmetrical. We again use an Amperian loop of radius r inside the wire, where $r < R$.

By symmetry, B is tangent to the loop, so Ampere's law gives:

$$B(2\pi r) = \mu_0 i_{\text{enc}}$$

The enclosed current is proportional to the area encircled by the loop:

$$i_{\text{enc}} = i \frac{\pi r^2}{\pi R^2} = i \frac{r^2}{R^2}$$

Substituting this into Ampere's law:

$$B(2\pi r) = \mu_0 i \frac{r^2}{R^2}$$

Solving for B :

$$B = \frac{\mu_0 i}{2\pi R^2} r$$

Thus, inside the wire, the magnetic field increases linearly with r and reaches its maximum at the surface of the wire ($r = R$), where it matches the field just outside the wire.

Problem 1:

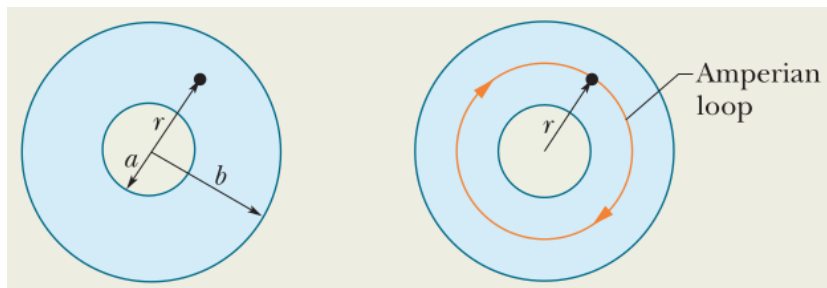


Figure shows the cross section of a long conducting cylinder has an inner radius $a = 2.0$ cm and an outer radius $b = 4.0$ cm. The current density inside the conductor is given by

$$J = cr^2,$$

where $c = 3.0 \times 10^6$ A/m⁴ and r is in meters. The total current flows out of the page.

Find the magnetic field B at a point located at $r = 3.0$ cm from the central axis of the cylinder.

Solution

Since the problem has cylindrical symmetry, we use Ampere's law:

$$\oint \mathbf{B} \cdot d\mathbf{s} = \mu_0 i_{\text{enc}},$$

where i_{enc} is the current enclosed by the Amperian loop of radius r .

To find i_{enc} , we integrate the current density over the cross-sectional area from the inner radius a to r :

$$i_{\text{enc}} = \int_a^r J dA.$$

The differential area element for a thin ring is given by:

$$dA = 2\pi r dr.$$

Substituting $J = cr^2$,

$$i_{\text{enc}} = \int_a^r cr^2(2\pi r dr).$$

Evaluating the integral,

$$i_{\text{enc}} = 2\pi c \int_a^r r^3 dr = 2\pi c \left[\frac{r^4}{4} \right]_a^r.$$

Substituting the limits:

$$i_{\text{enc}} = 2\pi c \left(\frac{r^4 - a^4}{4} \right).$$

Applying Ampere's law:

$$B(2\pi r) = \mu_0 i_{\text{enc}},$$

which gives

$$B = \frac{\mu_0}{2\pi r} \times 2\pi c \times \frac{r^4 - a^4}{4}.$$

Simplifying,

$$B = \frac{\mu_0 c}{4r} (r^4 - a^4).$$

Substituting values:

$$B = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(3.0 \times 10^6 \text{ A/m}^4)}{4(0.030 \text{ m})} \times [(0.030 \text{ m})^4 - (0.020 \text{ m})^4].$$

Calculating the values:

$$(0.030)^4 = 8.1 \times 10^{-6}, \quad (0.020)^4 = 1.6 \times 10^{-6},$$

$$B = \frac{(4\pi \times 10^{-7})(3.0 \times 10^6)}{4(0.030)} \times (8.1 - 1.6) \times 10^{-6}.$$

$$B = \frac{(4\pi \times 10^{-7})(3.0 \times 10^6)}{4(0.030)} \times 6.5 \times 10^{-6}.$$

$$B = 2.0 \times 10^{-5} \text{ T}.$$

Thus, the magnetic field at $r = 3.0 \text{ cm}$ is

$$\mathbf{B} = 2.0 \times 10^{-5} \text{ T}.$$

Magnetic Force on a Moving Charged Particle

The force acting on a charged particle moving through a magnetic field is given by the **Lorentz force**:

$$\mathbf{F}_B = q(\mathbf{v} \times \mathbf{B})$$

Explanation of Terms

- $\mathbf{F}_B$: Magnetic force acting on the particle (vector).
- q : Charge of the particle (scalar).
- $\mathbf{v}$: Velocity of the particle (vector).
- $\mathbf{B}$: Magnetic field (vector).
- $\times$: Cross product, indicating that the force is perpendicular to both the velocity and the magnetic field.

Direction of the Magnetic Force

The direction of $\mathbf{F}_B$ is determined by the **right-hand rule**:

1. Point your fingers in the direction of $\mathbf{v}$.
2. Curl them toward $\mathbf{B}$.
3. Your thumb points in the direction of $\mathbf{F}_B$ for a positive charge. For a negative charge, the force is in the opposite direction.

Magnitude of the Magnetic Force

The magnitude of the magnetic force is given by:

$$F_B = |q|vB \sin \theta$$

where:

- v : Speed of the particle.
- B : Magnitude of the magnetic field.
- θ : Angle between $\mathbf{v}$ and $\mathbf{B}$.

Key Observations

- If $\theta = 0^\circ$ or 180° (motion parallel or anti-parallel to $\mathbf{B}$), $F_B = 0$. The particle feels no magnetic force.
- The force is maximum when $\theta = 90^\circ$ (motion perpendicular to $\mathbf{B}$).

Problem 2:

A uniform magnetic field $\mathbf{B}$, with a magnitude of 1.2 mT, is directed vertically upward throughout the volume of a laboratory chamber. A proton with a kinetic energy of 5.3 MeV enters the chamber, moving horizontally. What magnetic deflecting force acts on the proton as it enters the chamber? Assume the mass of the proton is 1.67×10^{-27} kg. (Neglect Earth's magnetic field.)

Solution

The magnetic deflecting force acting on the proton is given by the Lorentz force:

$$F_B = qvB \sin \theta$$

where:

- $q = 1.6 \times 10^{-19}$ C: Charge of the proton.
- $B = 1.2$ mT $= 1.2 \times 10^{-3}$ T: Magnitude of the magnetic field.
- v : Speed of the proton, to be determined from its kinetic energy.
- $\theta = 90^\circ$: The angle between the velocity of the proton and the magnetic field (since the proton moves horizontally and the field is vertical).

The Speed of the Proton:

The kinetic energy of the proton is given by:

$$K = \frac{1}{2}mv^2$$

Rearranging for v :

$$v = \sqrt{\frac{2K}{m}}$$

Substituting the given values:

$$K = 5.3 \text{ MeV} = 5.3 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} = 8.48 \times 10^{-13} \text{ J}$$

$$m = 1.67 \times 10^{-27} \text{ kg}$$

$$v = \sqrt{\frac{2 \times 8.48 \times 10^{-13}}{1.67 \times 10^{-27}}}$$

$$v \approx 3.18 \times 10^7 \text{ m/s}$$

The Magnetic Force is:

$$F_B = qvB \sin \theta$$

Substitute $\sin \theta = 1$ (since $\theta = 90^\circ$):

$$F_B = (1.6 \times 10^{-19})(3.18 \times 10^7)(1.2 \times 10^{-3})$$

$$F_B \approx 6.10 \times 10^{-15} \text{ N}$$

Answer

The magnetic deflecting force acting on the proton as it enters the chamber is approximately:

$$F_B \approx 6.10 \times 10^{-15} \text{ N}$$

Faraday's Law of Induction and Two Experiments

Two Experiments

Let us examine two simple experiments to prepare for our discussion of Faraday's law of induction.

First Experiment:

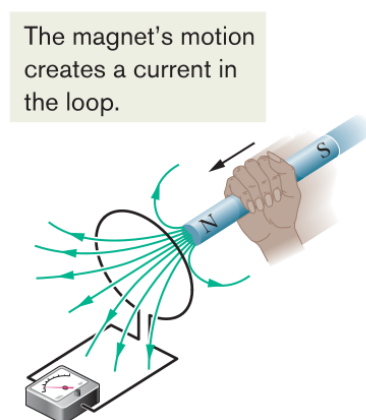


Figure 1: **An ammeter registers a current in the wire loop when the magnet is moving with respect to the loop**

Figure shows a conducting loop connected to a sensitive ammeter. Because there is no battery or other source of emf included, there is no current in the circuit. However, if we move a bar magnet toward the loop, a current suddenly appears in the circuit. The current disappears when the magnet stops. If we then move the magnet away, a current again suddenly appears, but now in the opposite direction. If we experimented for a while, we would discover the following:

1. A current appears only if there is relative motion between the loop and the magnet (one must move relative to the other); the current disappears when the relative motion between them ceases.
2. Faster motion produces a greater current.
3. If moving the magnet's north pole toward the loop causes, say, a clockwise current, then moving the north pole away causes a counterclockwise current. Moving the south pole toward or away from the loop also causes currents, but in the opposite directions.

Second Experiment:

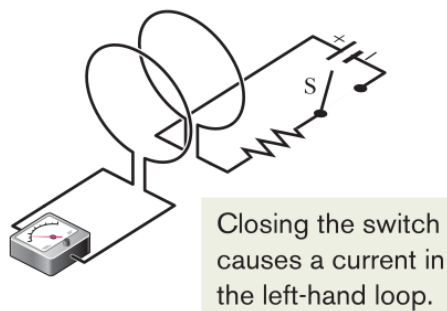


Figure 2: **An ammeter registers a current in the left-hand wire loop just as switch S is closed (to turn on the current in the right-hand wire loop) or opened (to turn off the current in the right-hand wire loop). No motion of the coils is involved.**

For this experiment, we use the apparatus of Figure 2, with the two conducting loops close to each other but not touching. If we close switch S to turn on a current in the right-hand loop, the meter suddenly and briefly registers a current—an induced current—in the left-hand loop. If we then open the switch, another sudden and brief induced current appears in the left-hand loop, but in the opposite direction. We get an induced current (and thus an induced emf) only when the current in the right-hand loop is changing (either turning on or turning off) and not when it is constant (even if it is large).

Faraday's Law of Induction

Faraday realized that an emf and a current can be induced in a loop, as in our two experiments, by changing the amount of magnetic field passing through the loop. He further realized that the “amount of magnetic field” can be visualized in terms of the magnetic field lines passing through the loop. Faraday's law of induction, stated in terms of our experiments, is:

An emf is induced in a loop when the number of magnetic field lines that pass through the loop is changing.

The current produced in the loop is called an *induced current*; the work done per unit charge to produce that current (to move the conduction electrons that constitute the current) is called an *induced emf*; and the process of producing the current and emf is called *induction*.

Lenz's Law

Lenz's law states that, **The direction of the induced current (and thus the induced emf) is always such that it opposes the change in magnetic flux that produced it.** Mathematically, Faraday's law of induction including Lenz's law is given by:

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

where $\mathcal{E}$ is the induced emf, and Φ_B is the magnetic flux given by:

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}.$$

If the loop is replaced by a closely packed coil of N turns, the induced emf is:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}.$$

The negative sign in Faraday's law represents Lenz's law, ensuring that the induced emf creates a current whose magnetic field opposes the change in flux.